

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 26

Topics covered :

1. Introduction to Linear Systems
2. Step and Impulse Response of a System

- In simple terms, a [signal](#) (often denoted as $x(t)$) is a function that carries some information. A [system](#), on the other hand, is a process that operates on a signal and produces another signal. Systems are often designed to process an input signal to generate a desired output signal. A system can be abstracted as a black box model, where the input $x(t)$ and output $y(t)$ are explicitly shown (as shown in Fig. 1). The black box represents all the mathematical operations the system performs to produce the desired output.

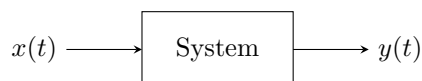


Fig. 1: Representation of a system

- For certain systems, there exists a simple mathematical relationship between the input and output, often referred to as the input-output relation. In this course, we will focus on systems where the input and output are related through a differential equation. Although we will primarily discuss systems modeled using differential equations, the theory can also be extended to systems modeled in other ways, such as with algebraic relations.
- A system is [linear](#) if it satisfies the principles of additivity and homogeneity. Specifically, if $y_1(t)$ and $y_2(t)$ denote the outputs of the system for inputs $x_1(t)$ and $x_2(t)$, respectively, then the system is linear if
 - the output of the system for the input $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$ (this property is called additivity).
 - the output of the system to the input $\alpha x_1(t)$ is $\alpha y_1(t) \forall \alpha \in \mathbb{C}$ (this property is called homogeneity).

The additivity and homogeneity properties can be combined into a single condition i.e., a system is linear if the output of the system for the input $\alpha x_1(t) + \beta x_2(t)$ is $\alpha y_1(t) + \beta y_2(t) \forall \alpha, \beta$. This condition is referred to as the [superposition principle](#).

- A system whose input-output relationship is governed by the relation

$$\left(\sum_{i=1}^n a_i(t) y^{(i)} \right) + a_0(t) y = x(t) \text{ i.e., } L[y] = x(t) \quad (1)$$

where $a_i(t)$ are polynomials of the independent variable (t) , is always [linear](#). To demonstrate this, let $y_1(t)$ and $y_2(t)$ denote the output (or response) of the system to inputs $x_1(t)$ and $x_2(t)$ i.e.,

$$y_1(t) = \{y : L[y] = x_1(t)\} \text{ and } y_2(t) = \{y : L[y] = x_2(t)\} \quad (2)$$

Now, we need to show that $\alpha y_1(t) + \beta y_2(t)$ satisfies the DE $L[y] = \alpha x_1(t) + \beta x_2(t) \forall \alpha, \beta \in \mathbb{C}$. This can be shown as follows:

$$L[\alpha y_1(t) + \beta y_2(t)] = \alpha L[y_1(t)] + \beta L[y_2(t)] = \alpha x_1(t) + \beta x_2(t) \quad (3)$$

Equation (3) shows that the output of the system to the input $\alpha x_1(t) + \beta x_2(t)$ is $\alpha y_1(t) + \beta y_2(t) \forall \alpha, \beta \in \mathbb{C}$, demonstrating that the system is linear.

- While we have covered methods for solving equation (1), we will now focus on the subclass of problems where the coefficients $a_i(t)$ are constants. In this context, so far we have covered:
 - Methods to find the solution (which represents the system's output) when the input $x(t)$ takes a specific form, either through the method of undetermined coefficients or variation of parameters.
 - Additionally, we've used the system's linearity property to analyze its response to periodic inputs, leveraging Fourier series to break down the input signals.

We'll now explore the responses of the system to other types of inputs, more specifically step and impulse inputs, which play a crucial role in the analysis and design of systems.

- One way of representing a system described by a differential equation is through a [block diagram](#). The block diagram of a system whose input-output relation is given by generic first- and second-order differential equations is shown in Fig. 2. The basic building block for a system whose input-output relation is governed by a differential equation is the integrator.

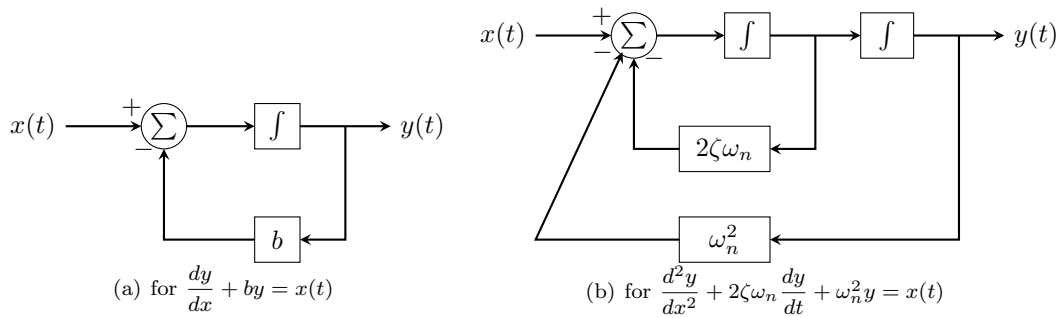


Fig. 2: Block diagram representation

- A [unit step signal](#) (shown in Fig. 3(a)) is defined as

$$u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases} \quad (4)$$

The steps involved in computing the response of the system to a shifted unit step function $u(t - \tau)$, referred to as the *step response* $\{y(t) : L[y] = u(t - \tau)\}$, are as follows:

1. **For $t < \tau$:** The input is 0, and the system's response is the solution of the homogeneous differential equation $L[y] = 0$. The arbitrary constants in the homogeneous solution can be determined using the initial conditions,

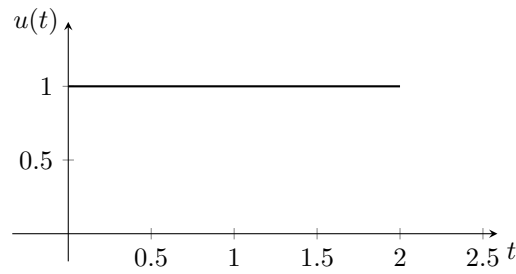


Fig. 3: Unit step signal

assuming that the initial conditions at t_0 are specified before the transition, i.e., $t_0 < \tau$. Let $y_1(t)$ denote the system's response for $t < \tau$.

2. **For $t \geq \tau$:** The input is 1, and the system's response (denoted as $y_2(t)$) is the solution of the non-homogeneous differential equation $L[y] = 1$. The arbitrary constants in the response $y_2(t)$ are determined using the initial condition $y_2(\tau) = y_1(\tau)$. This condition ensures that the overall response

$$y(t) = \begin{cases} y_1(t) & \text{for } t < \tau \\ y_2(t) & \text{for } t \geq \tau \end{cases} \quad (5)$$

is continuous, which is typically the case for most physical systems.